Research Directions for Variable Selection in Wasserstein Robust Testing

1 Starting point

The theory note writes the selection problem $\min_{|S|\le d} V(S)$ as the minimization of a convex function over $\mathcal{Z}_d$ and solves it by outer approximation. In development checks with the hinge loss, the method evaluated every subset before it closed the gap, for every cut we tried, and the relaxation was weak: at the uniform point $z = (d/D)\mathbf{1}$ it took the value 0.58, against the optimum 1.40 for $D = 10$, $d = 2$. This note argues that the weakness belongs to the problem, not to the formulation, and builds the research directions on a reformulation that makes this visible.

The argument runs in one line. Section 2 rewrites $V(S)$ as a divergence between the projected classes; for the hinge loss it becomes a budgeted matching. Section 3 uses the matching to explain the relaxation gap and the hardness. Section 4 turns the three consequences into a statistical, a computational and a game-theoretic direction, and Section 5 ranks them.

Notation follows the theory note. For a distribution Q on $\mathbb{R}^D$, $Q[S]$ is its image under $x \mapsto x[S]$, and $\mathcal{P}_k[S] = \{Q : \mathcal{W}_1(\widehat{\mathbb{P}}_k[S], Q) \le \rho_k\}$ is the projected ambiguity set, with the ℓ_1 cost on $\mathbb{R}^{|S|}$.

2 The robust value is a divergence between projections

For $a, b \ge 0$ let $\psi(a,b) = \inf_{t\in\mathbb{R}}\{a\ell(-t) + b\ell(t)\}$. It is concave and positively homogeneous, and $\psi(a,a) = 2a$ because convexity gives $\ell(-t) + \ell(t) \ge 2\ell(0) = 2$. Hence

$$D_\ell(Q_0, Q_1) = 2 - \int \psi(dQ_0, dQ_1)$$

is an f-divergence between probability distributions [7]. For $\ell(t) = (1+t)_+$, e^t and $\log_2(1+e^t)$ it equals $2\,\mathrm{TV}$, the squared Hellinger distance $\int(\sqrt{dQ_0} - \sqrt{dQ_1})^2$, and twice the Jensen–Shannon divergence in bits.

Proposition 2.1. *For every $S \subseteq [D]$,*

$$V(S) = 2 - \min_{Q_k \in \mathcal{P}_k[S]} D_\ell(Q_0, Q_1).$$

Proof. Fix Q_0 and Q_1. Minimizing the risk $\mathbb{E}_{Q_0}\ell(-f) + \mathbb{E}_{Q_1}\ell(f)$ point by point over $f \in \mathcal{F}_S$ gives $\int \psi(dQ_0[S], dQ_1[S])$. Projection maps the ambiguity sets into $\mathcal{P}_k[S]$, and every element of $\mathcal{P}_k[S]$ lifts with the same cost. Exchanging the minimum over f and the supremum over distributions therefore gives $V(S) \ge 2 - \inf D_\ell$, and Lemma 2.1 of the theory note attains this bound with distributions supported on the projected samples. □

Variable selection is thus the choice of d coordinates on which the two classes stay as far apart as possible, in the divergence chosen by the loss, after an adversary moves each class by ρ_k in $\mathcal{W}_1$. This places the problem next to methods that select a projection maximizing a distance between empirical distributions: sparse principal differences analysis [6], projected Wasserstein tests [13], feature-robust transport [9] and kernel variable selection [12]. Two things differ here: the distance is a divergence measured after a transport attack, and its value is the risk certificate of a minimax test. The mirror construction, a Wasserstein distance measured after a total-variation attack, is the robust transport of [8].

For the hinge loss the divergence is total variation, and the attack becomes combinatorial. Write $c_{ij}(S) = \|x_i^{(0)}[S] - x_j^{(1)}[S]\|_1$ and

$$m_S(\rho) = \max_{\pi\ge 0}\Big\{\sum_{i,j}\pi_{ij} :\ \sum_j \pi_{ij} \le \frac{1}{n_0},\ \sum_i \pi_{ij} \le \frac{1}{n_1},\ \sum_{i,j}\pi_{ij}c_{ij}(S) \le \rho\Big\}, \tag{1}$$

the largest cross-class mass that an adversary can pair up with total transport budget ρ.

Proposition 2.2 (Hinge loss)**.** $V(S) = 2\,m_S(\rho_0 + \rho_1)$.

Proof. By Proposition 2.1, $V(S)/2 = 1 - \min \mathrm{TV}(Q_0, Q_1)$ is the largest overlap $\int \min(dQ_0, dQ_1)$. Overlap at a point draws class-0 mass from some $x_i^{(0)}[S]$ and class-1 mass from some $x_j^{(1)}[S]$. Pairing the two, the triangle inequality charges at least $c_{ij}(S)$ per unit of overlap to the two budgets together, so the overlap is at most $m_S(\rho_0 + \rho_1)$. Conversely, let π be feasible in Eq. (1) with $\rho = \rho_0 + \rho_1$. Move the fraction $\theta = \rho_0/(\rho_0 + \rho_1)$ of each π_{ij} from $x_i^{(0)}$ to $x_j^{(1)}$, and the remaining fraction of it from $x_j^{(1)}$ to $x_i^{(0)}$. Class 0 spends at most $\theta\rho = \rho_0$, class 1 at most ρ_1, and the overlap is at least $\sum_{i,j} \pi_{ij}$. □

Three consequences follow, and the rest of the note uses them. First, the radii enter only through $\rho_0 + \rho_1$; on random instances with unequal radii the solver of the theory note agrees with $2m_S$ to 10^{-10}. Second, the least favorable distributions pair up cross-class points: the adversary is a matchmaker with a transport budget, and S separates the classes when few cross-class pairs are cheap in the coordinates of S. Third, evaluating $V(S)$ is a bipartite matching with one side constraint and $n_0 n_1$ variables, instead of a transport plan over N^2 pairs with two budgets.

3 Why convex relaxations are weak

Let $C_{ij}^m = |x_i^{(0)}[m] - x_j^{(1)}[m]|$, so that $c_{ij}(S) = \sum_{m \in S} C_{ij}^m$. Linear programming duality for Eq. (1) gives

$$m_S(\rho) = \min_{u,v,\lambda \geq 0} \Big\{ \rho\lambda + \bar{u} + \bar{v} : \; u_i + v_j + \lambda c_{ij}(S) \geq 1 \Big\}, \qquad \bar{u} = \frac{1}{n_0} \sum_i u_i, \;\; \bar{v} = \frac{1}{n_1} \sum_j v_j.$$

Each cross-class pair is either paid for by its endpoints or separated by its priced distance $\lambda c_{ij}(S)$, and selecting S decides which coordinates the adversary pays for. With $w_m = \lambda z_m$ the whole selection problem becomes one compact mixed-integer program, with $\rho = \rho_0 + \rho_1$:

$$\begin{aligned} \min_{z \in \mathcal{Z}_d,\, u,v,w,\lambda \geq 0} \quad & \rho\lambda + \bar{u} + \bar{v} \\ \text{subject to} \quad & u_i + v_j + \sum_m w_m C_{ij}^m \geq 1, \qquad w_m \leq \lambda, \qquad \rho w_m \leq z_m, \qquad \sum_m w_m \leq d\lambda. \end{aligned} \tag{2}$$

Its optimal value is $\min_{|S| \leq d} m_S(\rho) = \frac{1}{2} \min_{|S| \leq d} V(S)$: the point $u = \mathbf{1}$, $v = 0$, $\lambda = 0$ has value 1, so $\rho\lambda \leq 1$ at the optimum, and then $w_m = \lambda z_m$ is optimal.

The relaxation has a plain meaning. Replace $\mathcal{Z}_d$ by its convex hull $\{z \in [0,1]^D : \mathbf{1}^\top z \leq d\}$ and set $z = w/\lambda$. The relaxed value is the smallest matching value over weighted costs $\sum_m z_m |x[m] - y[m]|$: the relaxation lets the defender choose a weighted ℓ_1 metric instead of a set of coordinates. Weighted metrics are genuinely better defenses. Averaging coordinates concentrates the pairwise costs $\sum_m z_m C_{ij}^m$ around their mean, and the adversary loses the few cheap pairs it can always find in a sparse projection. Table 1 shows the effect: the relaxed value is one half to two thirds of the optimum. It is still stronger than the convex function of the theory note, which already takes the value 0.58 at the uniform point on the first instance. The gap therefore belongs to the problem: the relaxed problem is itself a sensible robust test, and it is better than every sparse one.

D	N	d	$\min_S V(S)$	relaxation of Eq. (2)	Eq. (2) by HiGHS	enumeration
10	30	2	1.40	0.89	0.2 s, 17 nodes	0.2 s
15	30	2	1.12	0.75	0.4 s, 25 nodes	0.5 s
12	30	3	0.87	0.49	0.7 s, 302 nodes	1.1 s
30	40	3	0.75	0.43	26 s, 4.1k nodes	22 s
50	40	3	0.80	0.40	235 s, 26k nodes	–
100	40	3	≤ 0.88	0.37	time limit 600 s	–

Table 1: Hinge loss, $\rho_0 + \rho_1 = 0.6$. Both classes are standard normal except that class 1 is shifted by 0.8 in coordinates 1–3 and scaled by 1.8 in coordinate 4. Program Eq. (2) reproduces the enumerated optimum wherever enumeration was run. Development check, not a formal result.

The combinatorial core is also hard in the worst case.

Proposition 3.1. *Let the data be binary and* $\rho_0+\rho_1 < \min\{1/n_0, 1/n_1\}$. *Then* $\min_{|S|\le d} V(S) \le 2(\rho_0+\rho_1)$ *if and only if some* S *with* $|S| \le d$ *separates every cross-class pair, that is,* $x_i^{(0)}[S] \neq x_j^{(1)}[S]$ *for all* i, j. *Deciding this is the* k*-feature-set problem, which is NP-complete [2] and W[2]-complete in* d *[1]. Hence computing* $\min_{|S|\le d} V(S)$ *is NP-hard.*

Proof. Write $\rho = \rho_0 + \rho_1$. If S separates every pair, then $c_{ij}(S) \ge 1$ for all pairs and $m_S(\rho) \le \rho$. Otherwise some pair has $c_{ij}(S) = 0$ and can be matched at no cost, so $m_S(\rho) \ge \min\{1/n_0, 1/n_1\} > \rho$. Proposition 2.2 completes the proof. □

Diminishing returns do not rescue greedy selection either. With class 0 at $(0,0), (1,1)$ and class 1 at $(0,1), (1,0)$, each coordinate alone leaves the projected classes identical, so $V = 2$, while the two coordinates together give $V = 2(\rho_0 + \rho_1)$ when $\rho_0 + \rho_1 \le 1$. A strong convex relaxation is therefore not to be expected. Progress must come from information that the weighted-metric problem does not see: the sample size, the matching structure, or the value of mixing coordinates.

4 Directions

4.1 Selection makes robust tests honest in high dimension

Table 2 draws both classes from the same distribution, so the correct robust value is 2. Yet $V([D])$ falls to 0.03 at $D = 50$. In high dimension the empirical points are isolated, every cross-class pair is expensive to match, and the robust test certifies a separation that does not exist. Selection limits the effect, but the best of many subsets is optimistic too, and in the first instance of Table 1 the optimal pair consists of two uninformative coordinates. A radius that ignores the dimension is not honest; an honest radius must depend on d and on the number of candidate subsets.

D	1	2	3	5	10	20	50
$V([D])$	1.98	1.66	1.17	0.59	0.21	0.09	0.03
$\min_{\lvert S\rvert=2} V(S)$	–	1.66	1.65	1.53	1.39	1.40	–

Table 2: Both classes $N(0, I_D)$, $n_0 = n_1 = 15$, hinge loss, $\rho_0 + \rho_1 = 0.6$; mean over five draws. Development check.

The detector certificate already contains what an honest radius needs.

Proposition 4.1. *Let* (g, a, λ) *solve the detector program at* S, *and let* f *be its extension in Appendix A of the theory note, used as the test that declares class* 1 *when* $f < 0$. *For any distributions* $\mathbb{P}_0, \mathbb{P}_1$ *on* $\mathbb{R}^D$ *with finite first moments,*

$$\mathbb{P}_0(f < 0) + \mathbb{P}_1(f \ge 0) \le \mathbb{E}_{\mathbb{P}_0}\ell(-f) + \mathbb{E}_{\mathbb{P}_1}\ell(f) \le V(S) + \sum_{k=0}^{1} \lambda_k\big(\mathcal{W}_1(\widehat{\mathbb{P}}_k[S], \mathbb{P}_k[S]) - \rho_k\big).$$

Proof. The first inequality holds because $\ell \ge 1$ on $[0, \infty)$. The envelope E_k of Appendix A dominates the class-k loss of f, is λ_k-Lipschitz for the ℓ_1 norm on $\mathbb{R}^{|S|}$, and satisfies $E_k(x_i^{(k)}[S]) \le a_i^{(k)}$. Kantorovich–Rubinstein duality gives $\mathbb{E}_{\mathbb{P}_k} E_k \le \frac{1}{n_k}\sum_i a_i^{(k)} + \lambda_k \mathcal{W}_1(\widehat{\mathbb{P}}_k[S], \mathbb{P}_k[S])$. Summing over k gives the second inequality. □

So $V(S)$ bounds the true total error as soon as each projected empirical distribution lies within ρ_k of the truth. Coverage is needed only in dimension $|S|$, and only for the subsets under consideration. For data in $[0,1]^D$, the rates of empirical measures [4, 14], a bounded-difference inequality and a union bound over the at most $(eD/d)^d$ subsets suggest a radius of order

$$\rho_k(d, \delta) = d\, n_k^{-1/d} + d\sqrt{\big(d\log(eD/d) + \log(1/\delta)\big)/n_k} \qquad (d \ge 3),$$

which makes the bound hold for all $|S| \le d$ at once with probability at least $1 - \delta$. Three results then seem within reach.

1. *Validity after selection.* The selected $\widehat{S}$ and its detector satisfy the error bound $V(\widehat{S})$ with no correction for the data-driven choice.

2. *A budget chosen by the certificate.* The bound is uniform over $d \le d_{\max}$, so $\widehat{d} = \arg\min_d \min_{|S|=d} V_{\rho(d)}(S)$ keeps a valid certificate. More coordinates separate the classes better but need larger radii, and the certificate balances the two.

3. *Recovery.* If $\mathbb{P}_0$ and $\mathbb{P}_1$ differ only on a set $S^\star$, with a margin in robust total variation, then $\widehat{S} = S^\star$ once n_k is large compared with the radius in dimension $|S^\star|$. The rate depends on d rather than D, as in the power analyses of [13, 12].

A sharper route replaces coverage by the variation-based analysis of Wasserstein robust optimization [5]; the Lipschitz constants λ_k in Proposition 4.1 play the role of the variation used there. A further question links back to Section 3: a dense weighted metric has a large effective dimension and would need a large honest radius, so with honest radii the weighted-metric relaxation may lose its advantage and become tighter. The story is that selection is not only for interpretability: it is what makes a Wasserstein robust test valid in high dimension.

4.2 Selection as matching interdiction

By Section 3, the defender prices d coordinates, and the adversary buys the largest cross-class matching it can afford. This is an interdiction game in which the defender raises costs and the follower solves a matching program; arc removal is the classical case [15]. The follower's feasible set shrinks as more coordinates are priced, which is close to the monotone setting of the interdiction cuts of [3].

The immediate step is to replace outer approximation by Eq. (2) for the hinge loss: it is exact, has $n_0 n_1$ rows and one price, and its relaxation is stronger than the current root. Table 1 also shows its limit. The search tree grows almost as fast as enumeration, as Section 3 predicts, so the remaining gains come from structure.

1. *Bounds from matchings.* A matching π stays feasible after scaling, so for every S

$$m_S(\rho) \ge \sum_{i,j} \pi_{ij} \cdot \min\Big\{1, \frac{\rho}{\sum_{m\in S} \kappa_m}\Big\}, \qquad \kappa_m = \sum_{i,j} \pi_{ij} C^m_{ij}.$$

   The bound uses the exact cost of π under S; the cuts of the theory note are weaker linear bounds of the same kind. Such bounds in the master problem, with the pair rows of Eq. (2) generated lazily, give the natural decomposition for large n: only pairs that are cheap in some candidate S matter.

2. *A statistical tolerance.* By Proposition 4.1, differences in V below the slack of the certificate carry no statistical meaning. Stopping the search at that tolerance turns exact certification into honest certification, and may stop early.

3. *General losses.* The same construction gives a compact program with two prices and N^2 rows. For the hinge loss, Proposition 2.2 reduces it to one price and $n_0 n_1$ rows.

4.3 Randomized selection

The relaxation gap measures how much mixing coordinates helps the defender. A randomized test collects this gain while each realization still reads only d coordinates. Let the defender draw a random pair (S, f) with $|S| \le d$ and $f \in \mathcal{F}_S$, and let the adversary choose $Q_k \in \mathcal{P}_k$ knowing the distribution of the pair but not its realization. The risk is linear in both strategies, so Sion's theorem [11], under the usual compactness conditions, gives

$$\min_{(S,f)\ \text{random}} \max_{Q_k\in\mathcal{P}_k} \mathbb{E}\big[\mathbb{E}_{Q_0}\ell(-f) + \mathbb{E}_{Q_1}\ell(f)\big] = \max_{Q_k\in\mathcal{P}_k} \min_{|S|\le d} \big[2 - D_\ell(Q_0[S], Q_1[S])\big].$$

The right side asks for one attack that confuses every d-coordinate projection at once, so the adversary must split its budget across coordinates. For example, take one point per class, $x^{(0)} = (0,0)$ and

$x^{(1)} = (1, 1)$, the hinge loss, $\rho_0 + \rho_1 = 1/2$ and $d = 1$. Each fixed coordinate gives $V = 1$. On the right side, an overlap o_m of the projections on coordinate m costs at least o_m, and the ℓ_1 cost adds over coordinates, so $o_1 + o_2 \le 1/2$ and the value $\min_m 2o_m$ is at most $1/2$: half the fixed-selection optimum.

Randomization is known to strictly help in adversarial classification [10]; here the exchange of minimum and maximum is exact, and the value is a divergence between projections. Two questions come first: whether the finite reduction of Lemma 2.1 survives in the mixed game, and how to compute the right side, for instance by generating subsets whose pricing problem is a selection for a fixed attack, with no inner maximization. The guarantee holds on average over the defender's coin, which not every application accepts.

5 Priorities

1. *Honest radii* (Section 4.1) offer the strongest story at the lowest cost. Proposition 4.1 is already proved; the next result is the uniform radius with a budget chosen by the certificate, and Table 2 is the motivating figure.

2. *Matching interdiction* (Section 4.2) starts with a direct implementation of Eq. (2) and a comparison on the 100-dimensional study, then adds the matching bounds and the statistical tolerance.

3. *Randomized selection* (Section 4.3) is exploratory; the finite reduction for the mixed game comes first.

Three lines are not pursued. Further cut engineering for outer approximation, such as cheap least-favorable cuts, Pareto cuts or several tangents per evaluation, reduced oracle time by 30–45% in development checks but cannot close a gap that belongs to the problem. Sinkhorn ambiguity sets place the selection inside $\exp(-c/\varepsilon)$ and lose the matching structure. Neural generators of least favorable distributions lose the certificate. All numbers in this note come from small synthetic development checks, not from formal experiments.

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